

HMAM NOORELDEEN ABDALLA ALTOOM

Gameplay & AI Systems Programmer | Unreal Engine 5

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PROFILE

Final-year Bachelor of Information Technology (Game Technology) student with hands-on experience designing adaptive AI systems, combat mechanics, and developer tools in Unreal Engine 5. Built a complete adaptive boss-AI tool as a Final Year Project, combining real-time player behaviour analysis with tunable combat systems. Seeking a Game Programming, Gameplay Programmer, AI Programmer, or Technical Designer internship on PC/console projects in Malaysia or Singapore.

EDUCATION

Bachelor of Information Technology (Game Technology) with Honours *Exp. Graduation: 2027*

Faculty of Information and Communication Technology, Universiti Teknikal Malaysia Melaka (UTeM)

- Currently completing Final Year Project: "Adaptive Combat Tool – A Boss Combat System in Unreal Engine 5"

KEY PROJECT

Adaptive Combat Tool – Final Year Project

Unreal Engine 5

Editor Utility Widget tool for adaptive boss AI design

- Designed and developed an Unreal Engine 5 Editor Utility Widget that lets developers build adaptive boss encounters without AI programming expertise.
- Implemented a real-time player intent classification system that tracks attacks, dodges, distance, and health to classify players as Aggressive, Defensive, Panic, or Balanced.
- Built four distinct boss behaviour modes (Normal, Counter, Chase Aggressive, Heavy Pressure), each with tunable combat parameters such as cooldown, range, movement speed, and damage.
- Developed both Fixed and Adaptive AI modes, giving developers the choice between manual behaviour selection and automatic player-driven adaptation.
- Created auto-setup and validation tools for integrating the system into existing player/boss actors and projects.
- Implemented supporting gameplay systems including light/heavy combo attacks, directional dodge, sprint, weapon hit detection, combo queueing, AI chasing/attacking, and animation-driven combat.

OTHER PROJECT EXPERIENCE

Whispers of the Peak – Team Project

Unreal Engine

First-person survival horror game (Workshop II Project)

- Contributed to gameplay design, combat and gameplay systems, core mechanics planning, and technical implementation as part of a development team.

TECHNICAL SKILLS

Game Development: Unreal Engine 5 (UE5.6) – Blueprint Visual Scripting (Advanced), C++ (Intermediate), AI Systems Design, Combat Systems, Gameplay Programming, Tool Development, Editor Utility Widgets, State Machines, Behaviour Systems, Combat Balancing, Level Design, Game Design Documentation, Git (Basic)

Environment & World Building: Landscape Creation, Terrain Generation, Gaea, Environment Design, Map Blockouts, Navigation Setup, Lighting Basics

Art & Content Creation: Blender, Basic Animation, Asset Integration, Character Setup, Animation Retargeting

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)

CAREER OBJECTIVE

- Seeking: Game Programming Internship, Gameplay Programmer, AI Programmer, or Technical Designer roles
- Platforms: PC and Console (PlayStation/Xbox)